

Exercise Sheet Week 7

Exercise 1. Let f be a function given by $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = ax^3 + b$. Find the derivative of f using differentiation from first principles.

Exercise 2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$

- (i) Which theorem from lectures allows us to say immediately that f is not differentiable at 0?
- (ii) Verify this by trying to find $f'(0)$ using differentiation from first principles.

Exercise 3. Find $h'(x)$ when h is given by

- (i) $h(x) = \sin(\sqrt{1 + \cos(x)})$
- (ii) $h(x) = \sec(\sqrt{\alpha x})$
- (iii) $h(x) = \cos(\cos(\cos(x)))$
- (iv) $h(x) = x^5 \sec\left(\frac{1}{x}\right)$
- (v) $h(x) = \sqrt{a + \sqrt{bx}}$
- (vi) $h(x) = \sin^2(3\sqrt{x})$

Exercise 4. Give an example of a function that is continuous everywhere but fails to be differentiable in two places. (Hint: think about how its graph might look and then think of a function compatible with it.)

Exercise 5. Using differentiation from first principles, find the derivatives of the following function rules

- (i) $f(x) = \cos(x)$
- (ii) $f(x) = \frac{1}{2x}$.

Exercise 6. (i) Given that the function f is differentiable at $x = a$, state the standard definition of the derivative $f'(x)$ at $x = a$.

(ii) Now consider the function

$$f(x) = \begin{cases} x^3 \cos\left(\frac{1}{x^2}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

- (a) Calculate $f'(x)$ for $x \neq 0$ and show that $\lim_{x \rightarrow 0} f'(x)$ does not exist.
- (b) Show by differentiation from first principles that f is differentiable at $x = 0$ and find the derivative of f there.
- (c) Use this to answer: If a function has a derivative defined at all points, is that derivative necessarily continuous?